Siva **Sathyaseelan D N**

Computational & Theoretical Chemistry · Scientific Machine Learning

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Education

Indian Institute of Technology (BHU), Varanasi

Varanasi, India

Integrated Dual Degree (B.Tech + M.Tech), Industrial Chemistry — *Nov. 2021 – 2026*

- CPI: 8.43 / 10
- **Master's Thesis:** "Applications of Machine Learning and Deep Learning in Predictive Analytics for Molecular Property Prediction, Materials Informatics, and Flavor Science", Advisor: Dr. V. Ramanathan.

Research Experience

Indian Institute of Technology (BHU), Varanasi

Varanasi, India

Research Assistant, Advisor: Dr. V. Ramanathan — *Jul. 2024 – Present*

- Predicted **semiconductor band gaps** from elemental composition, benchmarking Random Forest, XGBoost, and artificial and graph neural networks to publication-grade accuracy for materials-informatics screening.
- Built **multi-label and multi-class taste classifiers** that predict a molecule's flavour profile (sweet, bitter, sour, and more) from its chemical structure, pairing cheminformatics fingerprints with deep-learning models for flavour-science applications.
- Carried out *ab initio* electronic-structure studies of **sulfur-centred hydrogen-bond cooperativity** in ethane-1,2-dithiol clusters interacting with HF, H_2O, and NH_3, quantifying how cluster size strengthens these interactions.
- Investigated how **stereochemical ambiguity in SMILES** (chiral centres and E/Z isomerism) degrades molecular-property prediction.
- Built a **consensus chemometric pipeline** (PCA, PLS-DA, K-Means, VIP, SIMCA) over **full-spectrum Raman** data across three wavenumber ranges to authenticate 10 commercial ghee brands against reference fats, flagging lard adulteration and conclusively ruling out palm oil.

Varanasi, India

Teaching Assistant, Advisor: Dr. V. Ramanathan — *Jul. 2025 – May 2026*

- Led weekly **tutorial and problem-solving sessions** for **CHI-321** (Reaction Kinetics) and **CHI-313** (Quantum Chemistry, Group Theory & Molecular Spectroscopy), supporting a cohort of undergraduate students.
- Ran **laboratory sessions** for CHI-321 Reaction Kinetics, guiding students through experiments and data analysis and assessing their lab work.
- Mentored junior students in **Scientific Machine Learning**, machine learning, and deep learning through fundamentals, coding, and hands-on research projects.
- **Graded** assignments and quizzes and handled **invigilation duties** for examinations, with detailed feedback to strengthen conceptual understanding.

University of Cambridge, JuliaMolSim

Cambridge, UK (Remote)

Google Summer of Code, Advisor: Dr. Joe G. Greener — *May 2026 – Sept. 2026*

- Implemented **ANI-2x** and **E(3)-equivariant** (**NequIP**, **Allegro**) interatomic potentials natively in Julia for **Molly.jl**: ML-driven molecular dynamics with no Python runtime, matching TorchANI to $\sim 10^{-6}$ eV/Å.
- Derived **analytic forces** via custom backpropagation through the atomic-environment-vector (AEV) and per-element networks ($\sim$7–14$\times$ faster than autodiff).
- Wrote portable **GPU kernels** (KernelAbstractions: CUDA / Metal / ROCm) scaling to **15,000+ atom** proteins.
- **Benchmarked** against TorchANI, NNPOps, and bio-mlff; a write-reduced AEV kernel and on-device CSR made **Molly-CUDA the fastest** on the full 15,954-atom protein (energy **13 ms**, forces **31 ms**), up to **125**$\times$ over host CPU.
- **Exposure**: Julia, KernelAbstractions, CUDA/Metal, ML interatomic potentials, molecular dynamics.

Massachusetts Institute of Technology CSAIL, Julia Lab

Cambridge, USA (Remote)

SciML Fellowship, Advisors: Dr. Samuel A. Isaacson, Dr. Christopher Rackauckas — *May 2025 – Sept. 2025*

- Developed the exact **Gillespie SSA** and τ**-leaping** solvers (simple, explicit, implicit, and adaptive) for **JumpProcesses.jl**, writing hand-tuned **GPU kernels** for both that ran large trajectory ensembles up to **10**$\times$ faster than the CPU baseline.
- Built a **variable-rate jump** aggregator framework (VR-Direct / VR-FRM with constant-bound and forward-time modes) for state- and time-dependent rates coupled to ODEs, plus a direct event cache that removed redundant structures for improved memory efficiency.
- Contributed **GPU-optimized Poisson sampling** to **PoissonRandom.jl**, the random-number backbone for the τ-leaping solvers.
- Tuned the **exact vs. approximate** accuracy/speed trade-off, validated on SIR/SEIR epidemiological and chemical-reaction-network models.
- Enhanced SciML's exact and approximate solvers for **discrete stochastic models** and added reproducible **SciMLBenchmarks.jl** comparisons.
- **Exposure**: Julia, CUDA, kernel programming, scientific ML, stochastic simulation. [Report]

University of California, Santa Cruz OSPO

Santa Cruz, USA (Remote)

Summer of Reproducibility, Advisor: Dr. Ben Greenman — *May 2025 – Aug. 2025*

- Built a Python-based **AI agent** (**LLM fine-tuning** + tool integration) that detects **vulnerable type predicates** (code that silently breaks type guarantees) in open-source projects.
- Ran the agent over real-world codebases to catalogue these unsafe predicates into a labelled **testing / benchmarking** dataset.
- Designed a **type-narrowing benchmark** across Ruby, Static Python, Pyre, and Clojure to compare how gradual **type checkers** handle these cases.
- **Exposure**: Ruby, Python, Static Python, Pyre, Clojure, AI agents, LLM fine-tuning, Prompt engineering.

Publications

1. **S. Sathyaseelan D N**, Kesava Perumal D N, Bharti, V. Ramanathan*. "Predicting Band Gap from Chemical Composition: A Comparative Study of Machine Learning and Deep Learning Models." *Scientific Reports*, Nature Portfolio (2026).
2. Bharti, M. K. Tripathi, **D. N. Siva Sathyaseelan**, B. K. Mishra, V. Ramanathan. "Cluster-size-dependent cooperativity in sulfur-centred hydrogen bonds: ethane-1,2-dithiol with HF, H_2O, and NH_3." *Physical Chemistry Chemical Physics*, Royal Society of Chemistry, **28**(24), 14862–14871 (2026).
3. **D. N. Siva Sathyaseelan**, V. Ramanathan. "Predicting Molecular Taste: Multi-Label and Multi-Class Classification." *bioRxiv*, Cold Spring Harbor Laboratory, 2025.05.02.651828 (2025).
4. V. Sharma, **D. N. Siva Sathyaseelan**, A. K. Bind, V. Ramanathan. "Consensus Chemometric Evaluation of Commercial Ghee Authenticity Using Full-Spectrum Raman Fingerprinting." *(under review)*.
5. **S. Sathyaseelan D N**, Gaurav Singh, Anant Saxena, V. Ramanathan. "Stereochemical Ambiguity in SMILES: Impact on Machine Learning for Molecular Property Prediction." *(manuscript in preparation)*.

Work Experience

Ema (ema.ai)

Bangalore, India

Machine Learning Engineer — *May 2026 – Present*

- Built end-to-end **Temporal ML pipelines** (Auto-Learning & Continuous Learning) that auto-generate and refine LLM-based QA-rule instructions, running up to 120 concurrent LLM calls with convergence-based stopping, across ML engines, workflows, and observability.
- Built a self-hosted **live UI & API localization** engine (11 languages) with Helsinki-NLP OPUS-MT models on **CTranslate2**; removing the Python wrapper cut per-string latency 43 ms → ∼7 ms, keeping all data in-house with no external subprocessor.
- Designed **Prometheus observability** (ML-, LLM-, and data-quality metrics) and gRPC services across the Agent-QA platform.
- **Exposure**: Python, LLMs, Temporal, gRPC, GCP, Prometheus, MLOps, NLP.

Bangalore, India (Hybrid)

AI Resident — *Sep. 2025 – Apr. 2026*

- Developed a multilingual **PII detection & redaction** pipeline (Presidio + spaCy NER; GCP DLP with KMS crypto-deterministic de-identification; LLM audit) at **88%** accuracy; conversation-pairing cut LLM API calls by **67%** (478 transcripts).
- Improved the production **intent classifier** with chain-of-thought prompt restructuring (GPT-4.1 95.1→96.0%, GPT-4.1-mini 93.5→96.8%), upstream of all personas.
- Built a multi-agent **root-cause triaging** system for negative feedback (Wipro, Hitachi) at **75–80%** accuracy; shipped Wipro intent-classifier (96.8%) and tool-calling (92.1%) gains live.
- **Exposure**: Python, LLMs, agentic AI, prompt engineering, evaluation, NLP.

Validating Lightning Signer (Summer of Bitcoin 2025)

Remote

Software Engineering Intern — *May 2025 – Aug. 2025*

- Raised unit-test coverage of the **Validating Lightning Signer (VLS)**, a security-critical service that isolates Bitcoin Lightning signing keys, past **85%**, hardening the **Lightning Network**.
- Wrote **end-to-end and integration tests** and applied **cargo-mutants** mutation testing to catch bugs ordinary tests miss.
- Added **system-level tests** to the **GitLab CI/CD** pipeline so every change is validated in real time against a live **Core Lightning** node.
- **Exposure**: Rust, RPC, gRPC, CI/CD, Docker, Bitcoin Core, Core Lightning, Cryptography, Hashing, P2P Networks.

Cloud Native Computing Foundation (CNCF) (Linux Foundation Mentorship)

Remote

Software Engineering Intern — *Mar. 2024 – May 2024*

- Built condition-validation logic in **Go** for **Kyverno** (a Kubernetes policy engine) to verify container-image attestations before admission.
- Integrated **Cosign** and **Notary** to handle signed-image verification securely within **Kubernetes** admission control.
- Strengthened the **CI/CD pipeline** with automated test frameworks, improving the reliability of attestation validation across image sources.
- **Exposure**: Golang, Kubernetes, Kyverno, CI/CD, Cosign, Notary, Cloud Security, DevSecOps, Distributed Systems, Cluster Computing.

Open-Source Software

For a complete list of open-source software I have worked on, see my GitHub profile.

- </> **Molly.jl**, molecular-dynamics simulation in Julia; contributed a native ANI-2x ML interatomic potential — *Julia*
- </> **JumpProcesses.jl**, jump / stochastic simulation (SciML); variable-rate aggregators and GPU τ-leaping solvers — *Julia*
- </> **PoissonRandom.jl**, fast Poisson random sampling with GPU-optimized sampling for jump processes — *Julia*
- </> **SciMLBenchmarks.jl**, benchmarks for scientific-ML and differential-equation solvers — *Julia*
- </> **BandGapPredict**, ML/DL pipeline predicting semiconductor band gaps from chemical composition — *PyTorch, Python*

Honors, Awards & Fellowships

2025 **IIT Color Award**, Top 5 out of 9k+ students institute-wide for contributions in science & technology, IIT (BHU)
2025 **SciML Fellowship**, MIT CSAIL, Julia Lab · $3000
2026 **Google Summer of Code**, University of Cambridge, JuliaMolSim · $1500
2025 **Summer of Bitcoin**, Selected among the Top 68 interns of 45,000+ applicants · $3000
2025 **Summer of Reproducibility**, UC Santa Cruz OSPO · $1500
2024 **Linux Foundation Mentorship**, Selected among the Top 56 interns of 10,000+ applicants · $3000
2024 **Best Blind App & Sign Everything Prize**, ETHOnline · $1000

Achievements

2025 **Gold Medal, Table Tennis**, Spardha, IIT (BHU)
2024 **2× Gold & Bronze, Inter-IIT Tech Meet 13.0**, Across multiple problem statements, IIT (BHU)
2024 **Silver Medal, Table Tennis**, Spardha, IIT (BHU)
2024 **Bronze Medal, Table Tennis**, Udgosh, IIT Kanpur
2023 **Silver Medal, Inter-IIT Tech Meet 12.0**, Aptos Labs Challenge (team lead of 6)
2023 **Gold Medal, Table Tennis**, Spardha, IIT (BHU)
2022 **Gold Medal, Table Tennis**, Spardha, IIT (BHU)
2021 **Top 2% in Joint Entrance Examination (JEE) 2021**, Among 900k+ candidates appearing

Extracurricular Activity

Inter-IIT Tech Meet 13.0, IIT (BHU) Contingent — *Varanasi, India*

Deputy Contingent Leader — *2024*

- Led and coordinated the IIT (BHU) contingent of 90+ members across technical, managerial, and design problem statements.
- Oversaw team selection, logistics, and practice schedules; the contingent secured multiple podium finishes (2× Gold, Bronze).

Table Tennis Club, IIT (BHU) — *Varanasi, India*

Captain — *2024 – 2026*

- Captained the institute table-tennis team, representing IIT (BHU) at the home fest, out fest, and Inter-IIT Sports Meet.
- Organized regular training sessions and mentored junior players, contributing to Gold, Silver, and Bronze medal finishes.

Technical Skills

Languages — C++ (proficient); Python, Julia, Rust, Go, TypeScript/JavaScript, Java, SQL

ML / Scientific — PyTorch, scikit-learn, LLMs, NLP; SciML / DifferentialEquations.jl, JumpProcesses.jl, Molly.jl; DFT & electronic structure, molecular dynamics, cheminformatics (RDKit)

Systems / Tools — CUDA & GPU kernels, Docker, Kubernetes, Git, Linux, AWS, CI/CD, Node.js, React, Flask, GraphQL, LaTeX